Universal effective interactions of globular proteins close to liquid–liquid phase separation: corresponding-states behavior reflected in the structure factor
— Supplementary Material —

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Supplementary SAXS data

Figure S1(A) shows SAXS data on the form factor $P(Q)$ of dilute protein solutions ($c = 10$ mg/mL) with various amounts of additive as indicated (symbols). Data for $P(Q)$ were obtained from the $I(Q)$ data via Eq. (2) of the manuscript by subtracting the fitted background and assuming $S_{\text{eff}}(Q) = 1$. Despite the differences in additive concentration, the $P(Q)$ data follow the same Q dependence, indicating that the shape and size of the protein molecules remains unaltered. The data show a plateau at low Q and, as expected for a globular object, a minimum at high Q. In our previous work,[1] the scattered X-ray intensity of dilute lysozyme solutions has been described by the form factor of a prolate ellipsoid of revolution with a semi-minor axis of 16.0 Å and an axial ratio of 1.5. The $P(Q)$ based on the same model with the same parameters (solid line) also agrees with the present data. Hence, it is used here to minimize the number of free parameters in Eq. (2) of the manuscript. For completeness, it is interesting to note that the form factor calculated via the CRYSOL programme[2] (dotted line) also agrees with the data. Figure S1(B) shows the Q-dependence of the scattered intensity of a dilute protein solution at different temperatures. The SAXS data coincide within experimental uncertainty, indicating that $P(Q)$ does not depend on T over the relevant Q-range probed.

Figure S2 contains further scattering data of protein solutions (symbols) at various protein concentrations c, temperatures T and additive concentrations [GuHCl] as well as the corresponding model fits (lines). In each subfigure, the b_2 values resulting from the model fits are displayed as an inset. The b_2 from Figure S2(A,B) have been included in Figure 1(D) of the manuscript, those of Figure S2(C) in Figure 2(C) of the manuscript, and all results are displayed in Figure 3(B) of the manuscript.

Estimation of the critical temperature $T_{\mathbf{c}}$

The critical temperature T_{c} of experimental binodal lines can be estimated based on a critical scaling ansatz.[3,4] In the vicinity of the critical point, the concentration dependence of the binodal line is given by:

$$T(\phi) = T_{\text{c}} \left(1 - \alpha \left| \frac{c - c_{\text{c}}}{c_{\text{c}}} \right|^{1/\beta} \right) , \tag{S1}$$

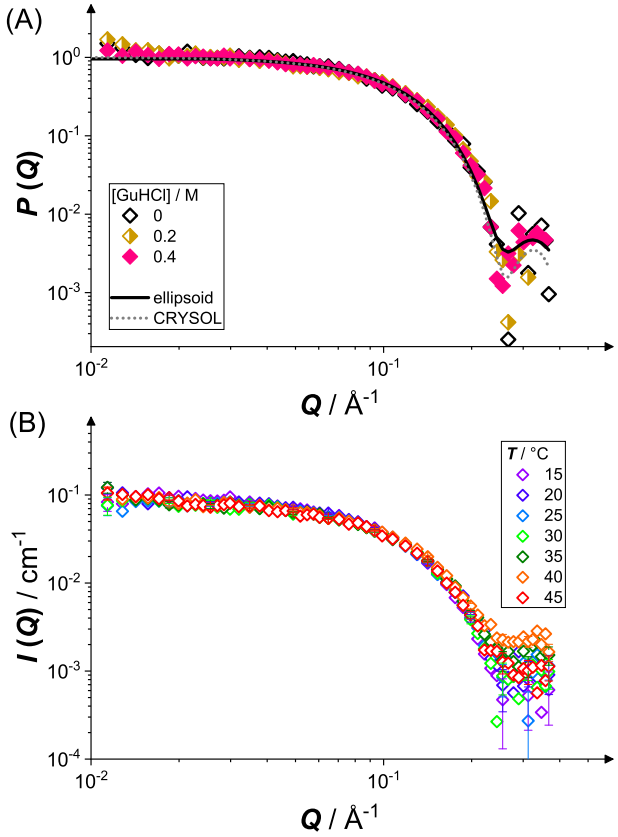

FIG. S1. (A) Form factor $P(Q)$ as a function of the magnitude of the scattering wavevector Q. Data (symbols) as obtained by SAXS experiments on dilute protein solutions (lysozyme, pH 4.5, 0.9 M NaCl, $c = 10$ mg/mL, $T = 20$ °C and guanidine concentration [GuHCl] as indicated) and fits (lines). (B) Scattered X-ray intensity $I(Q)$ as a function of the magnitude of the scattering vector Q of a dilute protein solution (lysozyme, pH 4.5, 0.9 M NaCl, $c = 10$ mg/mL) at different T (as indicated).

with the shape parameter α, the critical concentration c_c and the critical exponent for binary demixing $\beta = 0.325$, according to renormalization-group theory. The application of the critical scaling, Eq. (S1), to experimental binodal lines of protein solutions has been discussed in detail in our previous work.[5] The critical concentration of lysozyme binodals was found to vary between 200 and 300 mg/ml, depending on the solution conditions. Far away from the critical point, i.e. for very low and very high concentrations ($c \lesssim 70$ mg/ml and $c \gtrsim 500$ mg/ml, respectively), deviations between experimental binodal lines and the critical scaling were observed and, as a consequence, data at these concentrations were

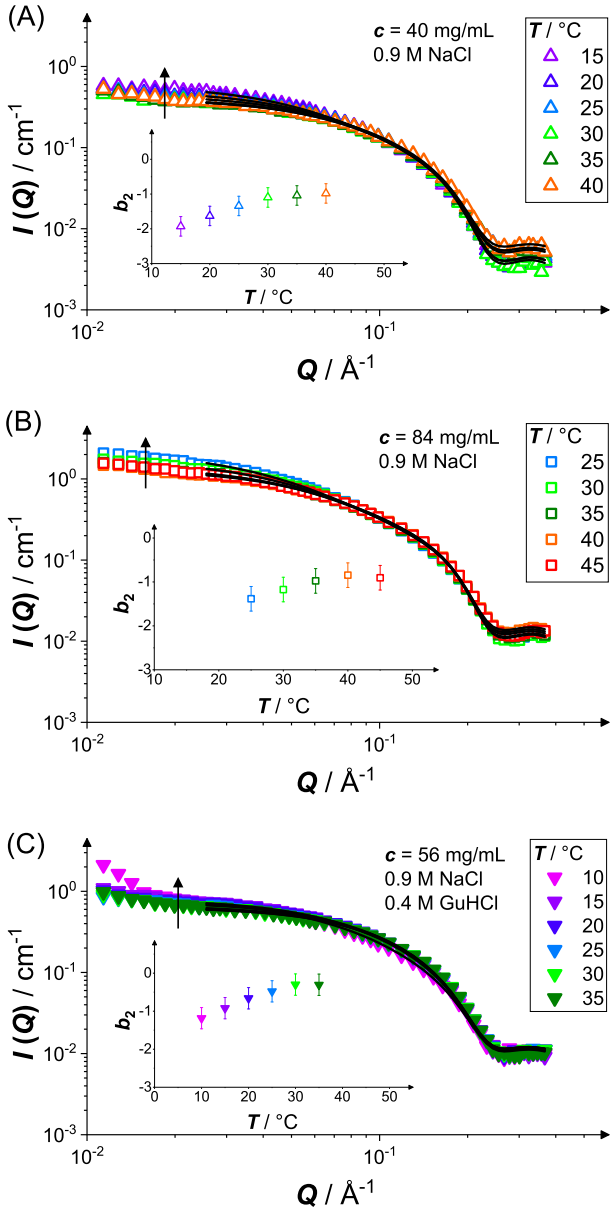

FIG. S2. Scattered X-ray intensity $I(Q)$ as a function of the magnitude of the scattering vector Q of protein solutions with fixed protein and additive concentrations (as indicated) at different temperatures T; data (symbols) and model fits (lines). The arrows indicate the approach to LLPS by lowering T. The insets show the normalized second virial coefficient b_2, as retrieved from the fits (colored symbols).

excluded from fits.

For the solution conditions shown in Fig. 1 of the manuscript, the binodal line has been

determined up to very high protein concentrations;[6] in Fig. S3 the data (grey symbols) is shown together with a fit of Eq. (S1) to the (constrained) data set (dashed line). Cloud-point data for the other solution conditions investigated here[7] (other symbols) are included as well. Assuming that the shape factor and the critical concentration are unaltered, a one-parameter fit (solid lines) to each (constrained) data set (crosses) yields estimates of T_c as presented in the inset of Fig. 3(A) of the manuscript with an uncertainty of approximately 2 K (which is comparable to the symbol size in the inset of Fig. 3(A) of the manuscript).

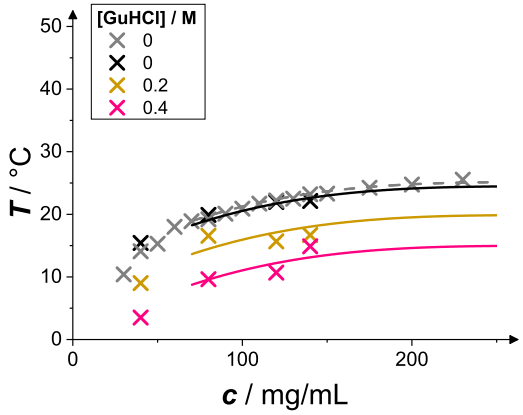

FIG. S3. Experimental binodal lines (symbols) of lysozyme solutions (pH 4.5, 0.9 M NaCl) in the presence of the various amounts of guanindine hydrochloride, as indicated.[6,7] Fits (solid lines) of Eq. (S1) to the data, yielding estimates of the critical temperature T_c, as shown in the inset of Fig. 3(A) of the manuscript.

[1] J. Hansen, R. Uthayakumar, J. S. Pedersen, S. U. Egelhaaf, and F. Platten, Interactions in protein solutions close to liquid–liquid phase separation: ethanol reduces attractions via changes of the dielectric solution properties, Phys. Chem. Chem. Phys. **23**, 22384 (2021).

[2] D. Svergun, C. Barberato, and M. H. J. Koch, CRYSOL: a program to evaluate x-ray solution scattering of biological macromolecules from atomic coordinates, J. Appl. Cryst. **28**, 768 (1995).

[3] M. L. Broide, T. M. Tominc, and M. D. Saxowsky, Using phase transitions to investigate the effect of salts on protein interactions, Phys. Rev. E **53**, 6325 (1996).

[4] M. Muschol and F. Rosenberger, Liquid–liquid phase separation in supersaturated lysozyme solutions and associated precipitate formation/crystallization, J. Chem. Phys. **107**, 1953 (1997).

[5] F. Platten, N. E. Valadez-Pérez, R. Castañeda-Priego, and S. U. Egelhaaf, Extended law of corresponding states for protein solutions, J. Chem. Phys. **142**, 174905 (2015).

[6] C. Gögelein, D. Wagner, F. Cardinaux, G. Nägele, and S. U. Egelhaaf, Effect of glycerol and dimethyl sulfoxide on the phase behavior of lysozyme: Theory and experiments, J. Chem. Phys. **136**, 015102 (2012).

[7] F. Platten, J. Hansen, J. Milius, D. Wagner, and S. U. Egelhaaf, Additivity of the specific effects of additives on protein phase behavior, J. Phys. Chem. B **119**, 14986 (2015).